

MAT3110/4110 Mandatory Assignment 1

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INTRODUCTION: PROBLEM 1.2.

Problem 1.2. is concerned with computing the solution to the two dimensional wave equation, which can be written as:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u, \quad (1)$$

PROBLEM 1.2.3.: EXACT SOLUTION

We need to show that the following complex function is a solution to the wave equation 1:

$$u(t, x, y) = e^{i(k_x x + k_y y - \omega t)} \quad (2)$$

To find this we find both the second derivative in time and the right hand side of the wave equation as follows:

$$\frac{\partial^2 u(t, x, y)}{\partial t^2} = -\omega^2 e^{i(k_x x + k_y y - \omega t)} \quad (3)$$

$$\nabla^2 u(t, x, y) = \frac{\partial^2 u(t, x, y)}{\partial x^2} + \frac{\partial^2 u(t, x, y)}{\partial y^2} = -k_x^2 e^{i(k_x x + k_y y - \omega t)} - k_y^2 e^{i(k_x x + k_y y - \omega t)} \quad (4)$$

These two expressions then need to be inserted into the wave equation and the expression needs to be simplified:

$$-\omega^2 e^{i(k_x x + k_y y - \omega t)} = c^2 \cdot \left(-k_x^2 e^{i(k_x x + k_y y - \omega t)} - k_y^2 e^{i(k_x x + k_y y - \omega t)} \right) \quad (5)$$

$$\omega^2 = c^2 (k_x^2 + k_y^2) \quad (6)$$

Therefore, eq. (2) solves eq. (1) as long as the following criterion is satisfied:

$$\omega = \pm c \sqrt{(k_x^2 + k_y^2)} \quad (7)$$

PROBLEM 1.2.4.: DISPERSION COEFFICIENT

The discretization of eq. (1) yields the following:

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{\Delta t^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} \right) \quad (8)$$

$$u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1} = \frac{c^2 \Delta t^2}{h^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n + u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n) \quad (9)$$

We now use the assumption that $m_x = m_y$, from which follows that $k_x = k_y = k$. Using this the statement from the chapter above can be refined to:

$$\omega = \pm \sqrt{2} ck \quad (10)$$

Furthermore, the solution to the discretized solution to eq. (8) is given as:

$$u_{ij}^n = e^{i(kh(i+j) - \tilde{\omega}n\Delta t)}. \quad (11)$$

This can be proven by inserting the solution into equation eq. (8), while also assuming that $CFL^2 = \frac{h^2}{c^2\Delta t^2} = 2$.

$$e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)} = \frac{1}{2} \cdot 2 \left(e^{i(kh(i+j+1) - \tilde{\omega}n\Delta t)} - 2e^{i(kh(i+j) - \tilde{\omega}n\Delta t)} + e^{i(kh(i+j-1) - \tilde{\omega}n\Delta t)} \right) \quad (12)$$

$$e^{-i\tilde{\omega}\Delta t} + e^{i\tilde{\omega}\Delta t} = (e^{ikh} + e^{-ikh}) \quad (13)$$

Which can be rewritten into:

$$2 \cos(\tilde{\omega}\Delta t) = 2 \cos(kh) \quad (14)$$

$$(\tilde{\omega}\Delta t) = kh \quad (15)$$

$$\tilde{\omega} = \frac{kh}{\Delta t} = \frac{\omega kh}{c\sqrt{2}} = \omega \quad (16)$$

For this we used $CFL = \frac{h}{c\Delta t} = \sqrt{2}$ and $k = \frac{\omega}{c\sqrt{2}}$, which was in turn derived from eq. (7).